

**Project Planning Phase**  
**Project Planning Template (Product Backlog, Sprint Planning, Stories, Story points)**

|               |                                                                   |
|---------------|-------------------------------------------------------------------|
| Date          | 29 June 2026                                                      |
| Team ID       | SWTID-2026-8752                                                   |
| Project Name  | Visualization Tool for Electric Vehicle Charge and Range Analysis |
| Maximum Marks | 5 Marks                                                           |

**Product Backlog, Sprint Schedule, and Estimation (4 Marks)**

Use the below template to create product backlog and sprint schedule

| Sprint   | Functional Requirement (Epic) | User Story No | User Story / Task                                                                                               | Story Points | Priority |
|----------|-------------------------------|---------------|-----------------------------------------------------------------------------------------------------------------|--------------|----------|
| Sprint-1 | Dataset Collection            | USN-1         | As a developer, I want to collect EV datasets from reliable sources so that accurate analysis can be performed. | 3            | High     |
| Sprint-1 | Data Cleaning                 | USN-2         | As a developer, I want to clean and preprocess datasets for accurate visualization.                             | 5            | High     |
| Sprint-1 | SQL Database                  | USN-3         | As a developer, I want to create SQL tables and import datasets for analysis.                                   | 3            | High     |
| Sprint-1 | SQL Queries                   | USN-4         | As a developer, I want to write SQL queries to analyze EV charging stations, brands, prices, and range.         | 5            | High     |
| Sprint-2 | Tableau Dashboard             | USN-5         | As a user, I want to view an interactive dashboard showing EV insights.                                         | 8            | High     |
| Sprint-2 | Charging Station Map          | USN-6         | As a user, I want to visualize charging stations across India.                                                  | 5            | High     |

|          |                     |        |                                                                                                                 |   |        |
|----------|---------------------|--------|-----------------------------------------------------------------------------------------------------------------|---|--------|
| Sprint-2 | Price Analysis      | USN-7  | As a user, I want to compare electric vehicle prices.                                                           | 3 | Medium |
| Sprint-2 | Range Analysis      | USN-8  | As a user, I want to compare driving ranges of electric vehicles.                                               | 3 | Medium |
| Sprint-3 | Brand Analysis      | USN-9  | As a user, I want to compare EV brands and available models.                                                    | 5 | High   |
| Sprint-3 | Efficiency Analysis | USN-10 | As a user, I want to identify the top efficient EV brands.                                                      | 5 | High   |
| Sprint-3 | Tableau Story       | USN-11 | As a user, I want to explore the dashboard through an interactive Tableau Story.                                | 5 | High   |
| Sprint-4 | Flask Website       | USN-12 | As a user, I want to access the Tableau Dashboard and Story through a web application.                          | 5 | High   |
| Sprint-4 | Documentation       | USN-13 | As a developer, I want to prepare project documentation and screenshots.                                        | 3 | Medium |
| Sprint-4 | GitHub Repository   | USN-14 | As a developer, I want to publish the complete project on GitHub for sharing and version control.               | 3 | High   |
| Sprint-4 | Final Testing       | USN-15 | As a developer, I want to test all dashboards, SQL queries, website pages, and documentation before submission. | 4 | High   |

### Velocity:

Imagine we have a 10-day sprint duration, and the velocity of the team is 20 (points per sprint). Let's calculate the team's average velocity (AV) per iteration unit (story points per day)

$$AV = \frac{\textit{sprint duration}}{\textit{velocity}} = \frac{20}{10} = 2$$

#### **Burndown Chart:**

A burn down chart is a graphical representation of work left to do versus time. It is often used in agile software development methodologies such as Scrum. However, burn down charts can be applied to any project containing measurable progress over time.

<https://www.visual-paradigm.com/scrum/scrum-burndown-chart/>

<https://www.atlassian.com/agile/tutorials/burndown-charts>

#### **Reference:**

<https://www.atlassian.com/agile/project-management>

<https://www.atlassian.com/agile/tutorials/how-to-do-scrum-with-jira-software>

<https://www.atlassian.com/agile/tutorials/epics>

<https://www.atlassian.com/agile/tutorials/sprints>

<https://www.atlassian.com/agile/project-management/estimation>

<https://www.atlassian.com/agile/tutorials/burndown-charts>